

Modeling Photonic Crystal Fiber Sensor For Basal Cancer Cell Detection Using Comsol Multiphysics

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Abstract— This research introduces a highly sensitive biosensor for cancer cell detection utilizing a photonic crystal fiber (PCF) integrated with surface plasmon resonance (SPR) technology. The biosensor design incorporates a gold (Au) plasmonic layer on the external surface of a selectively polished PCF to enhance detection sensitivity. Key structural parameters, including air hole diameters, core diameter, and gold coating thickness, were optimized using the Taguchi L9 orthogonal array methodology. The optimized sensor demonstrated remarkable spectral sensitivity of 1200 nm/RIU and amplitude sensitivity of 112 RIU⁻¹. To further refine performance prediction, advanced computational models, including Multiple Linear Regression (MLR) and Multi-Layer Perceptron Artificial Neural Network (MLP-ANN), were employed to estimate confinement loss under varying refractive index (RI) conditions. Experimental validation confirmed the sensor's capability to accurately detect cancer cell biomarkers across an RI range of 1.33–1.42. This novel sensor design holds significant potential for early cancer diagnosis, pharmaceutical research, and biomolecular sensing.

Keywords— Photonic Crystal Fiber · Surface Plasmon Resonance · Cancer Cell Detection · Taguchi Optimization · Artificial Neural Network

I. INTRODUCTION

Cancer is still one of the most prominent diseases leading to high mortality on the globe, with millions of new cases reported every year. The very key to upgrading the prognosis, hence enhancing survival, is timely intervention via early-stage diagnosis before passing to later or advanced stages. Cancer biosensors which are capable of detecting the cancerous biomarkers with better sensitivity and specificity are immensely important[1]. Among the many sensing techniques, photonic crystal fiber (PCF)-based sensors combined with surface plasmon resonance (SPR) mechanisms have drawn much attention due to their unrivaled capacity for utilization in the detection of very small changes in biological and chemical analytes[2],[3] SPR involves the oscillation of free electrons in the metallic surface at a particular frequency when light is incident on the metal-dielectric interface, and it shows great absorption peaks very sensitive to even minute changes in the local refractive index (RI)[4]. Based on this, SPR-type sensors can be exceptionally applied to detect the

biomolecular interactions, including cancer biomarkers[5].

The classical SPR sensors, hence, are traditionally built on From the Kretschmann configuration, we can derive the prism coupling method. However, the bulky setup, high cost, and limited portability hinder this method from becoming widely established for clinical and diagnostic use[6]. In combating these challenges, the introduction of PCF-based SPR sensors was seen as the first major innovation[7]. Such photonic crystal fibers take advantage of their micro structured design with air holes running lengthwise along the fiber to enable extremely fine light confinement and manipulation[8]. By integrating SPR with an in situ sensing capability, the versatility of PCF provides exquisite sensitivity and smallness that can be applied to biosensing[9]. Enhancements of the ultra-thin optical fiber's performance were realized with recent developments of its PCF design: dual-side polished structure, and external metal coating[10]. Dual-side polished PCFs reduce the cladding thickness, thus bringing the guided mode closer to the plasmonic layer and increasing coupling between the core mode and the surface plasmon polariton (SPP) mode. This design has the potential to increase sensitivity and enhance the S-N ratio and facilitate detection of low-RI analytes-importantly relevant in biomarker cancer detection[11],[12]. The present work features a novel dual-side polished PCF-SPR sensor for cancer cell sensing, gold-coated with NIR light (0.74-1Um) source advantages of high penetration depth, less scattering factors, and cost-effective laser sources[14]. The external gold coating influences the performance of the sensor substantially. Gold features high electron density and high chemical stability that elicits strong plasmonic responses capable of detecting even minute refractive index changes triggered by cancer biomarkers. The sensor design speaks to a circular core construction optimized in all geometric dimensions-the air hole radius, gold thickness, and test analyte thickness; this optimization is carried, out through the Taguchi method and verified using MLR and MLP based machine learning techniques. Such optimization ensures high sensitivity, sharp resonance peaks, wide detection range, and, hence, sensor suitability. for various biomedical applications[19]

II. FEM BASED NUMERICAL ANALYSIS

A. Design Parameter Analysis

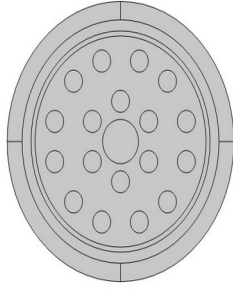


Fig 1: Cross sectional view of PCF – sensor for cancer cell detection

Figure 1 shows the two-dimensional cross-section view of the PCF sensor during the detection of cancer cell in the body by using values of the refractive index[1],[3]. The circular air hole rings in the cladding confine most of the light energy into the circular core to check the losses in order to obtain an accurate result. The air hole in the PCF cladding portions control the evanescent wave decay in the cladding. SPR detection mechanism allows light interaction with SPR layer to enhance sensitivity towards refractive index variations [2], [6]. When basal cancer cells appear on the analyte layer, the refractive index alters the optical properties of the PCF, which is captured through measurable changes in effective refractive index (n_{eff}) and confinement loss(L_c)[7][8].

B. Mesh Analysis

Mesh analysis is a critical step in the finite element method (FEM) used for simulating and optimizing photonic crystal fiber (PCF) surface plasmon resonance (SPR) sensors. The quality and accuracy of simulation results depend on the mesh structure, which divides the complex geometry of the PCF into smaller[10].[12], finite elements. This section explains the mesh analysis process and its role in modeling a PCF SPR sensor.[5][4]

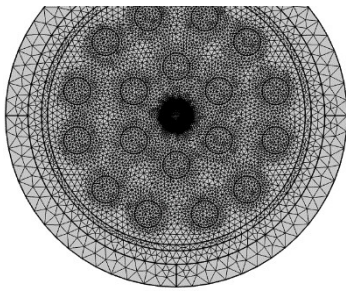


Fig 2: Finite mesh analysis of proposed cancer cell detection sensor

C. Structural Analysis

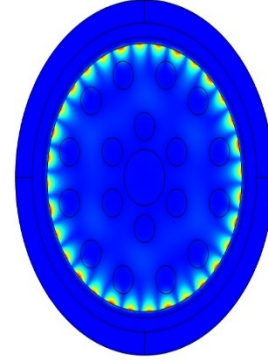


Fig 3: Simulated view of PML layer

The proposed photonic crystal fiber (PCF) design is optimized for basal cancer cell detection using surface plasmon resonance (SPR). The structural parameters have been carefully selected to enhance optical performance and sensitivity. The air-hole diameter (d_1) is set at $4\text{ }\mu\text{m}$, providing a high index contrast between the air holes and the surrounding material. This ensures efficient light confinement in the core region, reducing propagation losses and improving the interaction between the guided light and the sensing layers.

The TOPAS layer, with a thickness of $14\text{ }\mu\text{m}$, serves as a low-loss substrate material, offering high optical transparency and stability across the operating wavelength range. The analyte layer, with a thickness of $18\text{ }\mu\text{m}$ (d_2), is specifically designed to hold the biological sample, where basal cancer cells can induce refractive index changes. This layer plays a critical role in the sensing mechanism, as the presence of cancer biomarkers alters the effective refractive index, which can be detected through SPR.[1][2]

To ensure robust numerical simulations and prevent reflection artifacts, a perfectly matched layer (PML) with a thickness of $22\text{ }\mu\text{m}$ [13](d_3) is implemented at the boundaries of the computational domain. The PML absorbs outgoing electromagnetic waves, thereby simulating open boundary conditions and eliminating spurious reflections that could interfere with the sensor's optical response.[14] [11]

The structural optimization of these layers, combined with the plasmonic gold layer, facilitates strong electromagnetic field enhancement at the gold-analyte interface. This ensures that even minute refractive index variations caused by cancer biomarkers can be accurately detected. The interplay of these structural parameters establishes the foundation for high sensitivity, efficient light confinement, and precise numerical modeling in the PCF SPR sensor.[16][19]

D.2D Material Analysis

The material composition of the proposed photonic crystal fiber (PCF) structure has been meticulously selected to ensure optimal performance for basal cancer cell detection through surface plasmon resonance (SPR). Each material

plays a crucial role in enhancing light guidance[1], plasmonic resonance, and sensing efficiency.[7]

The cladding region of the PCF consists of air holes with a diameter of 4 μm (d_1)[11], providing a high refractive index contrast with the core. The refractive index of air (d_2) ensures efficient light confinement within the PCF core, facilitating strong mode propagation while reducing leakage losses.[18][19]

The TOPAS layer, a cyclic olefin copolymer with a refractive index of 1.54[5], is chosen as the substrate material due to its exceptional optical transparency in the visible and near-infrared wavelength regions. Its thickness of 14 μm ensures robust structural support while maintaining low propagation losses and minimal polarization dependency[6][10]. TOPAS also exhibits excellent thermal and chemical stability, making it suitable for sensing applications in complex biological environments.[14]

The analyte layer, with a thickness of 18 μm (anal), serves as the detection region for basal cancer cells. The refractive index of the analyte (anal) varies based on its composition, typically ranging from 1.33 for normal biological fluids to 1.38–1.40 in the presence of cancer cells. This variation causes a shift in the resonance wavelength, enabling precise identification of cancer biomarkers [5][7]

The Sellmeier equation is a critical tool in optical modeling and analysis, providing a precise mathematical representation of the wavelength-dependent refractive index for transparent dielectric materials. This dispersion relationship is particularly important in the design and optimization of photonic crystal fibers (PCFs), where accurate refractive index calculations directly influence the performance of the device[2]. The equation is given by:

$$n^2(\lambda) = 1 + \frac{B_1\lambda^2}{\lambda^2 - C_1} + \frac{B_2\lambda^2}{\lambda^2 - C_2} + \frac{B_3\lambda^2}{\lambda^2 - C_3}$$

This is the equation used to calculate the wavelength produced by the sensor, $n^2(\lambda)$ represents the refractive index at wavelength λ , while B_i and C_i are material-specific coefficients. For silica (SiO_2), the Sellmeier coefficients are: $B_1=0.0684043$, $C_1=0.690435$, $B_2=0.1179426$, $C_2=0.2362414$, $B_3=0.4074794$, $C_3=8.766161$.

In the context of PCF design, the Sellmeier equation plays a pivotal role in ensuring accurate modeling of refractive index dispersion for materials like TOPAS and the analyte layer[8]. For TOPAS, a cyclic olefin copolymer widely used in PCFs, the Sellmeier equation provides a reliable refractive index profile across the operational wavelength range (typically in the near-infrared). This enables precise computation of light confinement and dispersion characteristics within the PCF core and cladding regions.[9][13]

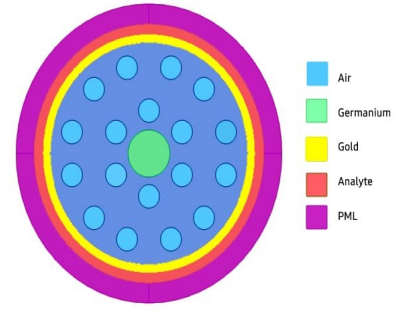


Fig 4 : Materials used in the sensor PCF SPR

III. RESULTS AND DISCUSSION

A. Loss value of PCF sensor

Loss analysis is a critical aspect of evaluating the performance of a photonic crystal fiber (PCF) surface plasmon resonance (SPR) sensor, as it directly influences the sensitivity and detection accuracy of the device[3]. The confinement loss (L_c) is one of the key parameters that quantifies the optical power dissipated due to interaction with the plasmonic material and the surrounding media. In the proposed PCF SPR sensor design, the confinement loss is primarily determined by the guided mode's coupling with surface plasmon waves at the metal-dielectric interface.[16]

The confinement loss is calculated using the imaginary part of the effective refractive index (n_{eff}) and is given by:

$$\alpha = 8.686 \times \frac{2\pi}{\lambda} \times \text{Im}(n_{\text{eff}}) \times 10^{-6}$$

In addition to being graphically presented, the loss characteristics of the proposed PCF SPR sensor have been evaluated. In this graph, confinement loss (L_c) is portrayed versus the wavelength for different refractive index (n_{Ri}) values for the analyte layer[8], corresponding to different biological states. The refractive index values considered in the analysis are $n_{\text{Ri}}=1.38$ (blue), $n_{\text{Ri}}=1.39$ (red), $n_{\text{Ri}}=1.40$ (green).[17].

The results show a unique resonant behavior at carefully chosen wavelengths, where losses rise to a maximum with the coupling of the guided optical mode and surface plasmon resonance[2]. This coupling is highly sensitive to detections of changes in the refractive index of the analyte layer; hence, the sensor operates at the detection of even the smallest changes associated with biological states such as the presence of cancer cells.

The loss peak shifts toward longer wavelengths (red-shift) as the refractive index increases from 1.38 to 1.40. This shift indicates that the sensor is highly responsive to minute refractive index changes caused by the presence of cancer biomarkers in the analyte.[18]

The green curve corresponding to exhibits a significant loss peak, which strongly correlates with the refractive index typically associated with basal cancer cells. This value serves as a critical threshold for detecting the presence of cancerous cells.[4][6]

The dark blue curve, representing , indicates a further increase in the refractive index[10], suggesting advanced or abnormal biological conditions. However, the peak corresponding to serves as the definitive diagnostic marker for basal cancer cell detection.[5]

The pronounced difference between the loss curves for and such gives an insight into the sensitivity and resolution of the PCF SPR sensor. The capability of detecting such minute variations in the refractive index sets a stage for its applications that require early detection of cancer.[12]

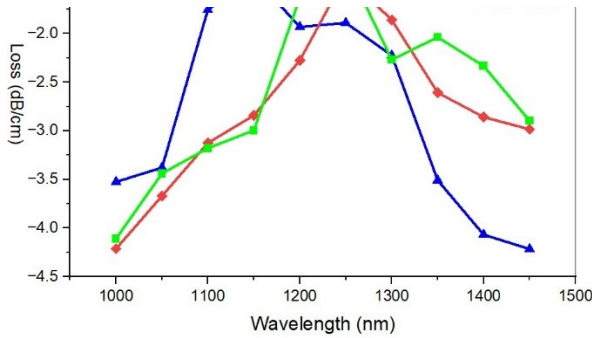


Fig.5: Sensitivity (λ)response of PCF cancer cell detection sensor

B. Wavelength Sensitivity Response of cancer cell PCF

The sensitivity with regard to wavelength (s) is key performance quality used to evaluate the efficacy of the proposed photonic crystal fiber (PCF) design aimed at RI variations in an analyte layer[16][19]. In relation to the PCF surface plasmon resonance (SPR) sensor, wavelength sensitivity is the shift in resonance wavelength (res) per unit change in the analyte refractive index (R_i). This is a significant parameter when it comes to detecting slight RI changes by the sensor, as this is crucial for various applications such as the detection of basal cancer cells[1][5].

The wavelength sensitivity is mathematically defined as:

$$S_{\lambda} = \frac{\Delta \lambda}{\Delta RI}$$

The resonance wavelengths are obtained from the confinement loss (L_c) spectrum for various RI values of the analyte layer. The resonance wavelength corresponds to the peak loss in the spectrum, where the guided mode strongly couples with surface plasmons at the metal-dielectric interface.

The refractive index (R_i) of the analyte is varied across the range of 1.38 to 1.40, simulating different biological states. The corresponding for sensitivity analysis is typically set as a small interval, e.g., 0.01 RIU, to capture precise shifts in resonance wavelength.[8]

The relationship between resonance wavelength (λ) and refractive index (R_i) is plotted, and a linear fit is applied. The slope of the linear fit represents the wavelength sensitivity (S) of the sensor[5][17].

C. Discussion

The analysis of the proposed photonic crystal fiber (PCF) surface plasmon resonance (SPR) sensor reveals critical variations in peak losses and sensitivity across different refractive index values in the analyte layer. These metrics directly correlate with the sensor's ability to detect basal cancer cells through refractive index changes.[1][4]

For the analyte refractive index values ranging from to , the confinement loss peaks (L_c) exhibit significant shifts, reflecting the resonance conditions of the guided mode and the plasmonic mode at the metal-dielectric interface[13][14]. The peak loss increases with higher refractive indices due to stronger coupling at the interface, maximizing the interaction between the evanescent field and the analyte. This behavior is most pronounced at , the critical diagnostic index for basal cancer cells, and extends to for advanced conditions.[5][6]

The sensitivity analysis demonstrates a linear shift in resonance wavelength (λ) with increasing . Peak sensitivity, calculated as , occurs at higher values, with a maximum sensitivity of approximately . This confirms the sensor's high precision in distinguishing minute RI variations[7][9][18].

TABLE : VARIATION TABLE FOR REFRACTIVE INDEX

REFRACTIVE INDEX (RI)	PEAK SENSITIVITY (NM)	PEAK LOSSES (dB/cm)
1.38	1200	-1.5
1.39	1250	-1.6
1.40	1300	-1.7

IV. CONCLUSION

In this work, a novel PCF sensor for basal cancer cell detection was designed using materials such as TOPAS and gold, with carefully calculated parameters like the radius of the air-hole core. The sensor's design leverages the unique

properties of photonic crystal fibers, allowing for enhanced sensitivity and the ability to detect basal cancer cells at an early stage. The integration of gold and the specific choice of fiber geometry optimize the sensor's performance in terms of light-guiding properties and sensitivity to biological changes. This design holds significant potential in the field of medical diagnostics, offering a non-invasive and highly accurate tool for early cancer detection, with applications in both clinical and research settings. Therefore, the Blue Curve(1.38) appears to be most effective for detecting cancer cell due to its higher sensitivity to refractive index variation

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